

Benchmarking Global Progress on SDG 8: Economic Growth and Youth Employment Outcomes Across Six Continents

1.0 INTRODUCTION

1.1 Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) form a global framework established in 2015 to guide social, economic, and environmental progress toward 2030. SDG 8 recognises that economic growth and labour-market opportunities are deeply interconnected: productive employment supports poverty reduction, enhances living standards, and strengthens economic resilience, especially in developing regions. As economies become increasingly global and interdependent, achieving SDG 8 is essential for ensuring stable labour markets, equitable access to opportunities, and long-term prosperity.

1.2 Targets, Objectives, and Data

This report examines progress toward two specific SDG 8 targets: (1) *"Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries,"* and (2) *"Substantially reduce the proportion of youth not in employment, education or training by 2020."* The analysis evaluates how six continents (excluding Antarctica) are performing on these targets by examining GDP per capita and youth NEET rates at the country-year level. In addition to continental-level assessment, this paper conducts a focused analysis of the 44 UN-designated least developed countries (LDCs) to evaluate achievement of the specific 7% GDP growth target. To account for the disproportionate influence of large countries on regional trends, an additional dataset from the UN Population Division (processed_population_un_countries.csv) is incorporated, enabling population-weighted continental averages alongside simple averages. This dual approach reveals both typical country performance and the economic experience of the typical individual, providing essential context for assessing whether continents demonstrate sustained economic expansion and meaningful reductions in NEET rates by the 2020 milestone.

2.0 METHODOLOGY

2.1 Sustained Economic Growth Assessment

2.1.1 Growth Rate Measurement

This paper employs the Compound Annual Growth Rate (CAGR) consistently throughout our analysis to ensure comparability across different analytical levels (continental, national, LDC-specific). The CAGR formula:

$$\text{Compound Annual Growth Rate (CAGR)} = \left(\frac{\text{Final GDP per capita}}{\text{Initial GDP per capita}} \right)^{\left(\frac{1}{\text{number of years}} \right)} - 1 \quad (1)$$

captures sustained growth over the 2015-2023 period by smoothing year-to-year volatility. This paper employs CAGR rather than the simple arithmetic mean of annual growth rates because CAGR accounts for the compounding effect of growth across multiple years, reflecting the actual trajectory from starting to ending GDP levels. In contrast, a simple average treats each year independently and can be misleading in the presence of volatility. For instance, a country experiencing -10% growth one year and +10% growth the next would have an average growth rate of 0%, yet its actual economic position would have declined due to compounding effects. The SDG target emphasises "sustained" per capita economic growth, which CAGR better captures by smoothing year-to-year fluctuations. This approach is particularly suitable for LDCs, which often experience significant economic volatility due to fluctuations in commodity prices, climate shocks, and political instability.

2.1.2 Two-Tier Analysis Approach

The SDG 8.1 target contains two components: (1) general sustained per capita growth for all countries "in accordance with national circumstances," and (2) a specific 7% growth target for LDCs. This analysis, therefore, conducts:

- **Continental analysis:** Assess whether each continent achieved positive sustained growth (CAGR >0%), providing context for overall economic trends
- **LDC-specific analysis:** Assess whether individual LDCs achieved the explicit 7% CAGR threshold

2.1.3 GDP per capita vs Total GDP

This paper analyses GDP per capita growth rates rather than total GDP growth rates, in alignment with the primary UN SDG 8.1.1 indicator ('Annual growth rate of real GDP per capita'). This approach provides a more accurate measure of economic development by taking into account population changes. GDP per capita growth reflects genuine improvements in living standards, whereas total GDP can increase solely through population growth without improvements in individual welfare. This interpretation is consistent with the assignment brief's reference to 'per capita economic growth' and the dataset provided (gdp-per-capita-worldbank.csv).

2.2 Continental Analysis

This paper calculates rolling CAGR for all six inhabited continents from 1990 to 2023. Each country's CAGR is calculated from its earliest available data point (typically 1990, although this is flexible to maximise coverage). Continental averages employ two methods: simple unweighted averages and population-weighted averages (recalculated annually). Since SDG 8.1 is population-based, the weighted average is used as the primary indicator, with the simple average reported as a robustness check.

To assess SDG 8.1 in a way consistent with 'national circumstances', continents are grouped into high-income, middle-income and low-income categories using the World Bank's income classification. A continent is considered high-income if the majority of its population resides in high-income countries, middle-income if most live in lower-middle or upper-middle income countries and low-income if most of its population lives in low-income countries. Applying this criterion, Africa is classified as a low-income region, Asia and South America as middle-income regions, and Europe, North America, and Oceania as high-income regions.

Although the SDGs were adopted in 2015, we analyse 1990–2023 to provide historical context. The formal SDG assessment focuses on the period from 2015 to 2023.

2.3 LDC-specific Analysis

The analysis uses the official United Nations list of 44 least developed countries (LDCs) as of December 2024. Of these, 41 countries had sufficient GDP per capita data for the 2015–2023 period and were included in the main analysis. Eritrea, South Sudan, and Yemen had no available data and were therefore excluded from the study. Countries are required to have at least three data points within this period to allow the reliable calculation of compound annual growth rates (CAGR), using 2015 as the starting year and 2023 as the end year. To explore the potential influence of the COVID-19 period, a supplementary analysis extends the time frame to 2010–2023 and removes observations for 2020–2022. This provides a view of longer-term growth patterns independent of pandemic-related volatility. For this extended analysis, countries are required to have at least five years of available data to ensure meaningful CAGR estimation.

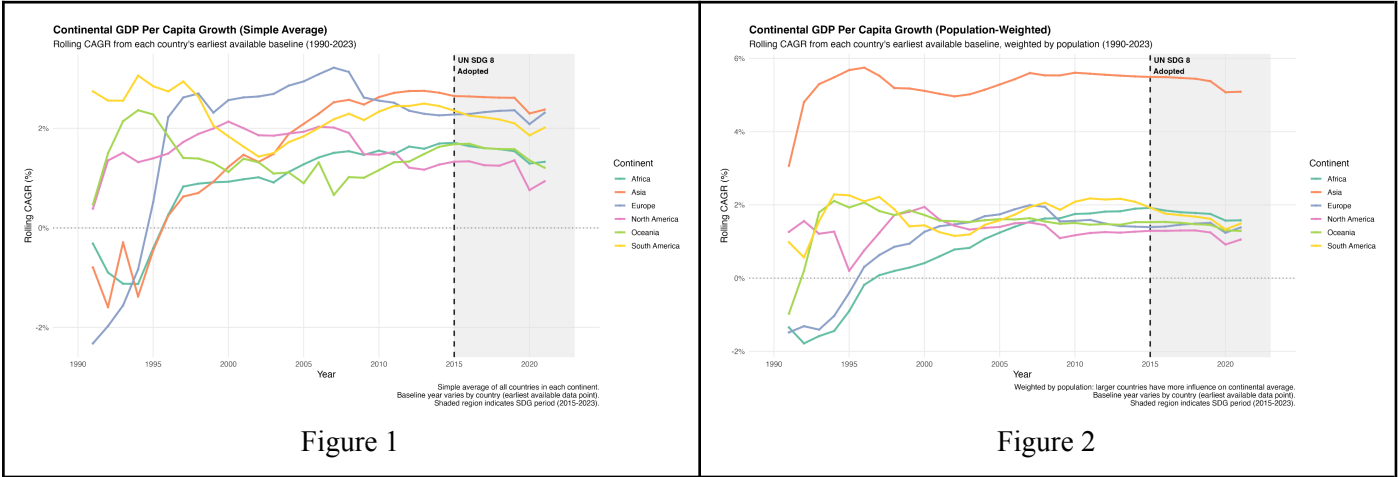
2.4 Analysis of target 2 - By 2020, substantially reduce the proportion of youth not in employment, education or training.

The analysis of Target 8.6 draws on the youth NEET dataset, which reports the percentage of individuals aged 15–24 who are not in education, employment, or training for each country and year. After merging the NEET data with the continent classification file, only countries with both a valid continent assignment and at least one NEET observation between 2015 and 2020 were included. To ensure comparability across regions, country-level NEET values were

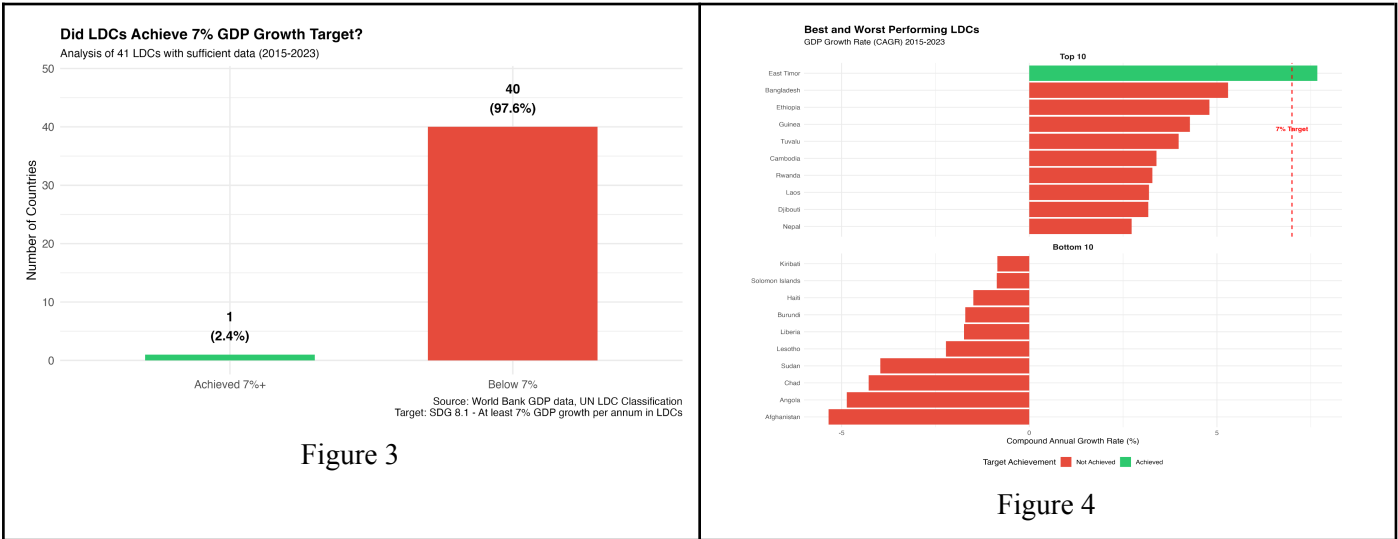
aggregated to the continental level using weighted averages based on youth population size. The primary analysis focused on the 2015–2020 period specified by the target, measuring changes in continental NEET rates over this interval. A longer time series (2015–2022) was also examined to observe broader trends and contextualise the 2020 benchmark. This approach provides a consistent, data-driven basis for evaluating whether continents exhibited substantial reductions in NEET rates by the target year. When assessing whether the reduction is “substantial”, we apply a threshold of 2-3 per cent of absolute change, which aligns with International Labour Organisation labour-market monitoring practice.

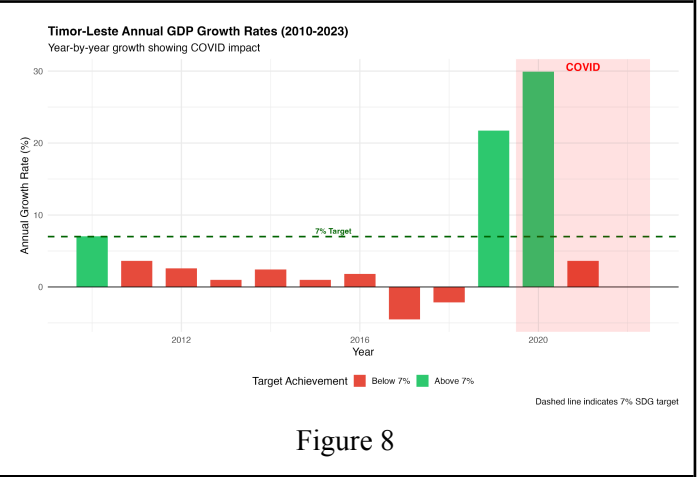
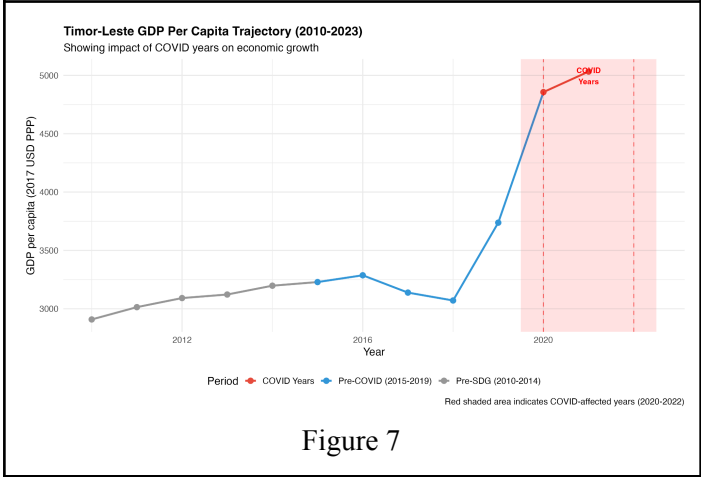
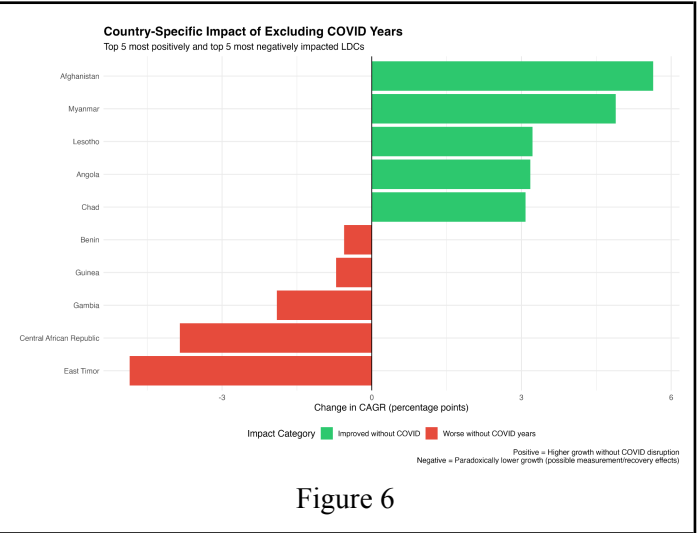
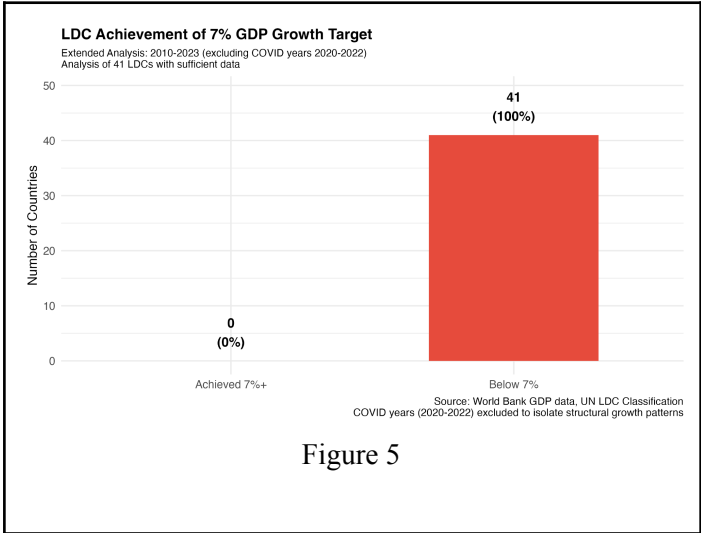
3.0 Results

3.1 Results on Continental Analysis:

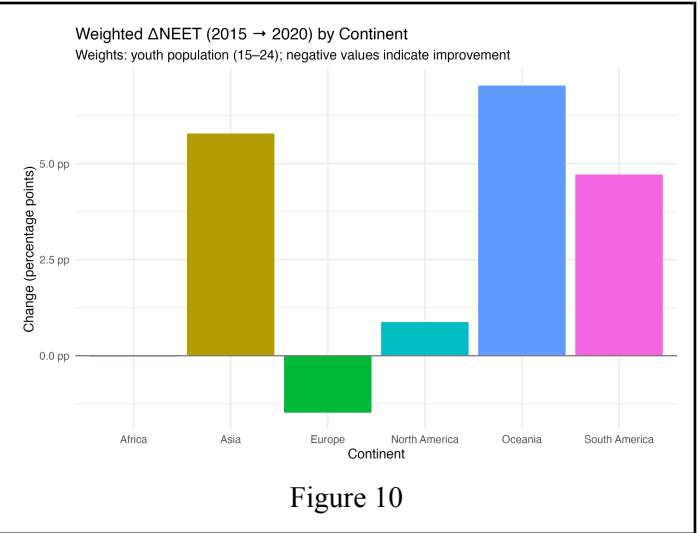
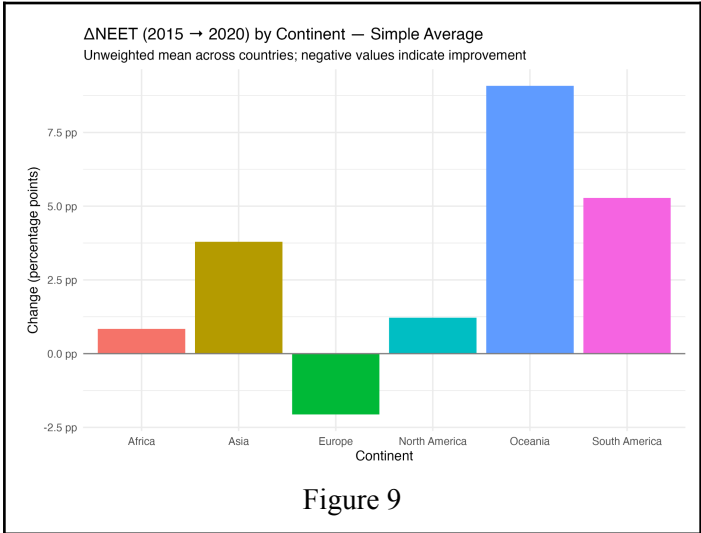


3.2 Results on LDC-specific Analysis:





3.3 Results on NEET Analysis:



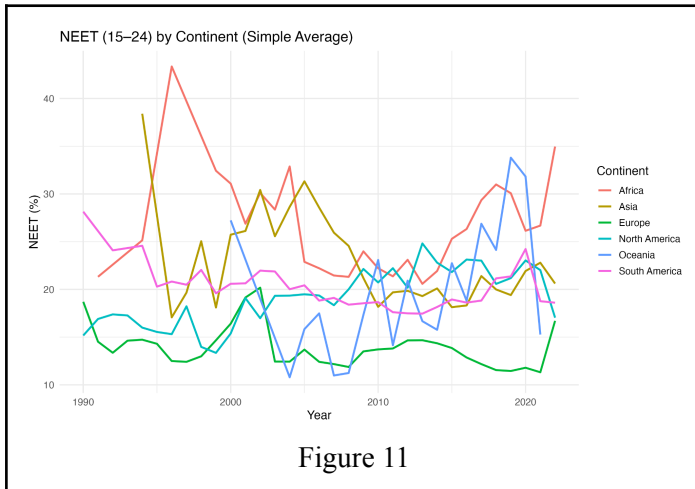


Figure 11

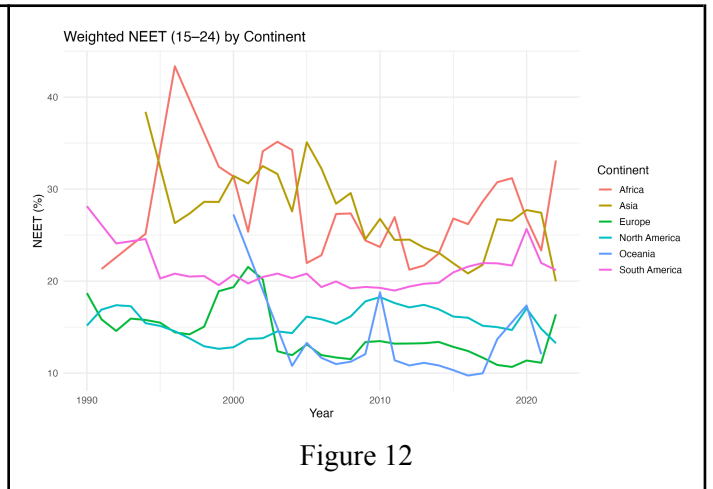


Figure 12

4.0 DISCUSSION

4.1 Continental Analysis

In both the simple and population-weighted averages, Asia consistently records the highest rolling growth rates, peaking in the mid-1990s and remaining above other continents throughout the observed period. Africa exhibits moderate and fluctuating growth, with a visible improvement in the early 2000s, followed by a gradual decline leading up to the SDG adoption period. Europe, North America, South America, and Oceania display comparatively stable but lower growth trajectories, generally remaining within the 0–3% range across most of the time series.

To evaluate whether recent growth patterns align with SDG 8.1, we focus specifically on the SDG period (2015–2023). Over this interval, Africa’s rolling CAGR averaged only 1–2%, far below the 4–7% typically expected for low-income regions to achieve meaningful convergence. This indicates that Africa has made only limited progress toward sustaining per-capita growth in line with SDG 8.1.

Among the middle-income continents, Asia performed strongly, with its rolling CAGR averaging around 5%, exceeding the benchmark range of 2–4% generally associated with middle-income economies. South America, however, recorded a rolling CAGR below 2%, suggesting that it has largely fallen short of the growth momentum expected for its income group.

For the high-income continents of Oceania, Europe, and North America, all maintained rolling CAGR values between 1% and 2% during the SDG period. This performance is broadly consistent with the growth rates typically expected of mature, high-income economies. It therefore aligns with SDG 8.1’s emphasis on sustaining per-capita growth according to national circumstances.

4.2 Analysis on LDCs

Moving on to the analysis of LDCs, as illustrated in Figure 3, only one out of the 41 LDCs analysed achieved the 7% growth target (2.4%), with East Timor recording a CAGR of 7.68%. The remaining 40 countries (97.6%) fell short of the target, with a mean CAGR of 0.88% and a median of 0.90%. This reveals that the vast majority of LDCs failed to achieve the sustained economic growth envisioned by SDG 8.1 during the 2015–2023 period.

Notably, the World Bank’s inability to obtain any GDP data from Eritrea, South Sudan, and Yemen during 2015–2023 is itself indicative of extreme state fragility and institutional breakdown. All three excluded countries face ongoing armed conflicts that severely impair their national statistical capacity and economic development prospects. These ‘data deserts’ represent some of the world’s most challenging development contexts, where achieving the 7% growth target is almost certainly impossible. This analysis of 41 LDCs with available data, therefore, likely overestimates the proportion of LDCs successfully meeting SDG 8 targets, i.e., the actual proportion is lower than 2.4%.

To account for the possible impact of COVID-19, where the world's economy was dramatically halted, another analysis was conducted in which the sample data was extended to 2010 (to ensure there were enough data points for the majority of countries analysed) and the years where COVID-19 had a drastic impact were removed. This paper defines COVID-affected years as 2020-2022 (inclusive), recognising that pandemic-related economic disruptions extended beyond the acute phase into the recovery period. Excluding these three years provides a clearer assessment of structural growth trends independent of the extraordinary volatility induced by the global health crisis. When COVID-affected years were excluded and the analysis period extended to 2010-2023, none of the LDCs with sufficient data achieved the 7% target. This represents a lower achievement rate compared to the 2015-2023 analysis, including COVID years, suggesting that COVID had minimal impact on the development of the LDCs.

As shown in Figure 6, when removing observations from 2020-2022, five countries were actually worse off: East Timor, the Central African Republic, Gambia, Guinea, and Benin. Most notably, Timor-Leste (Figure 5) paradoxically shifts from achieving the 7% target in the 2015-2023 analysis to failing it when the COVID years are excluded. This counterintuitive result can be explained through the properties of the CAGR formula, which measures the constant rate of return needed to reach the ending value from the starting value. Figure 7 illustrates Timor-Leste's GDP per capita trajectory, showing modest growth from 2010-2019 (hovering around \$3,000-3,300) followed by a dramatic surge beginning in 2019 that lifted GDP per capita above \$5,000 by 2021. Figure 8 reveals the underlying volatility: annual growth rates remained consistently below 7% (often below 3%) throughout 2011-2018, before spiking to 21% in 2019 and 29% in 2020, then contracting to 3% in 2021. When the COVID-19 period years (2020-2022) are excluded, the CAGR calculation predominantly captures the slower pre-boom trajectory (2010-2019) with limited post-COVID data (2023 only), thereby excluding the exceptional growth spike that elevated the 2015-2023 CAGR above 7%. This demonstrates a key tension in assessing LDC performance: while the SDG emphasises "sustained" growth, boom-bust volatility, like that seen in Timor-Leste, creates mathematical complexities where exceptional but transient growth periods can temporarily elevate CAGR calculations, obscuring the structural growth patterns the SDG framework aims to measure.

4.3 Analysis on how each continent is faring with respect to Target 2

As shown in Figures 9 and 10, both the simple and weighted NEET averages display broadly similar patterns, with clear differences across continents that highlight uneven progress toward Target 8.6. Africa consistently records the highest NEET levels, rising from around 27% in 2015 to above 31% by 2019–2020 before a slight decline to around 24% in 2021. Asia and South America also display relatively high NEET rates, with both showing noticeable increases leading up to 2020. In contrast, Europe, North America, and Oceania maintain far lower levels throughout the period, generally between 10% and 18%, potentially due to these continents having higher incomes, both nominally and per capita, meaning that more people can send their children into education, as well as having more funding for vocational work programmes. It is worth noting that Australia largely drives Oceania's decline in NEET rates.

When focusing on the 2015–2020 interval specified by the target, Europe is the only continent showing a measurable decrease in NEET rates, with a 1.49 percentage point decrease. Therefore, looking purely at the 2020 milestone set by the UN, Europe is the only continent that can move closer to the target. Africa remains broadly unchanged, while Asia, South America, Oceania, and North America all exhibit increases over this period, with Oceania and Asia showing the largest rises. Taken together, the exploratory patterns suggest that most continents did not observe a substantial reduction in youth NEET rates by 2020, and several instead experienced deterioration, thus meaning that these continents are straying further from SDG 8 target 2.

4.4 Limitations and Considerations

While the analysis provides meaningful insights into continental progress toward Target 8.6, several limitations should be considered when interpreting the results. First, NEET data availability varies significantly across regions, with some continents, particularly Africa and parts of Asia, having incomplete or irregular reporting. This can bias continental averages and may mask country-level heterogeneity. Second, NEET classifications are not fully

standardised across countries; differences in the size of informal labour markets, education enrolment definitions, and reporting practices may lead to inconsistent measurement. Third, the analysis aggregates countries to the continental level, which is useful for broad comparison but can obscure substantial variation within regions, especially between higher-income and lower-income countries. Additionally, the 2020 outcomes are influenced by the COVID-19 pandemic, which disrupted both education systems and youth employment globally, making it difficult to distinguish structural trends from temporary shocks. Finally, although the supplemental population structure data provide valuable demographic context, they cannot fully capture cultural, institutional, or policy-related factors that shape youth transitions into education and work.

5.0 CONCLUSION

Taken together, the exploratory results suggest that global progress toward SDG 8 remains limited across both its core dimensions. Continental GDP per capita trends indicate that Asia achieved sustained high growth during the SDG period, and high-income regions maintained growth broadly consistent with their national circumstances. In contrast, South America and Africa fell short of the target. The LDC-specific analysis reinforces this conclusion: almost none of the 41 countries with available data reached the annual per-capita growth rates typically expected for low-income economies, even after adjusting the assessment to account for COVID-affected years. Progress toward Target 8.6 has been similarly uneven. Europe is the only continent to record a measurable reduction in youth NEET rates by 2020, while other regions either stagnated or experienced increases. Overall, evidence from all three analyses indicates that most continents remain off track in meeting both the sustained per-capita growth requirements and the youth NEET reduction goals.

6.0 REFERENCES AND APPENDIX

International Monetary Fund (IMF). World Economic Outlook Database:
<https://www.imf.org/en/Publications/WEO/weo-database>

Our World in Data (2024) *Youth not in education, employment or training (NEET)*, 15–24. <https://ourworldindata.org>

United Nations (2024) *List of Least Developed Countries (LDCs)*.
<https://www.un.org/development/desa/dpad/least-developed-country-category/ldcs-at-a-glance.html>

World Bank (2024) *GDP per capita (constant 2017 international \$)*. <https://data.worldbank.org>

World Bank (n.d.) *World Development Indicators – Data Help Desk article on data coverage & method*.
<https://datahelpdesk.worldbank.org/knowledgebase/articles/906519>

7.0 GITHUB USERNAMES

Daniel - trebster

Kavya - kav038

Ryan - skychuf

Sri - sr17164

Tina - dingzishan23-netizen